Font-End Technologies (CSS)

BIS605

Software Development Technologies for Digital Business

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Resources

- https://www.w3schools.com
- https://www.geeksforgeeks.org
- https://www.w3docs.com/
- https://www.tutorialspoint.com/html

CSS

- Stands for Cascading Style Sheets
- It is used as simple mechanism for adding style to web documents.
- It describes how HTML elements are displayed on screen.
- It can control layout of multiple web pages all at once.

CSS Syntax

- A CSS consists of a selector and a declaration block.
- The selector indicates HTML element.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS property name and a value, separated by a colon.

CSS Syntax

Selector

Declaration

```css
p {
   color: red;
   text-align: center;
}
```

Property

Value

```html
<p>Hello World!</p>
<p>These paragraphs are styled with CSS.</p>
```

Hello World!

These paragraphs are styled with CSS.

CSS Selector

- Selector is used to find HTML elements which we want to style.
- There are 5 categories of selectors
  - Simple selector – based on name, id, class
  - Combinator selector – based on specific relationship between elements
  - Pseudo-class selectors – based on a certain state
  - Pseudo-elements selectors- select a part of an element
  - Attribute selectors – based on an attribute or attribute vale

CSS Simple Selector

- Simple selector will select element based on its id, and class attributes.
- To select id write a hash (#) followed by the id of element.

```
<style>
#para1 {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<p id="para1">Hello World!</p>
<p>This paragraph is not affected by
the style.</p>

</body>
```

Hello World!

This paragraph is not affected by the style.

CSS Simple Selector

- To select class write a period (.) followed by the class name.
- An element can also refer to more than one class.

```
<style>
.center {
  text-align: center;
  color: blue;
}

.large {
  font-size: 300%;
}
</style>
</head>
<body>

<h1 class="center">This heading will
not be affected</h1>
<p class="center">This paragraph will
be red and center-aligned.</p>
<p class="center large">This paragraph
will be red, center-aligned, and in a
large font-size.</p>

</body>
```

This heading will not be affected

This paragraph will be red and center-aligned.

This paragraph will be red, center-aligned, and in a large font-size.

CSS Universal Selector

- The universal selector (*) selects all HTML elements

```html
<style>
* {
  text-align: center;
  color: blue;
}
</style>
</head>
<body>

<h1>Hello world!</h1>

<p>Every element on the page will
be affected by the style.</p>
<p id="para1">Me too!</p>
<p>And me!</p>

</body>
```

CSS Grouping Selector

```css
h1 {
  text-align: center;
  color: red;
}

h2 {
  text-align: center;
  color: red;
}

p {
  text-align: center;
  color: red;
}
```

```css
h1, h2, p {
  text-align: center;
  color: red;
}
```

CSS Combinator Selector

- Combinator selector selects elements based on their relationship.
  - descendant selector (space)
  - child selector (>)
  - adjacent sibling selector (+)
  - general sibling selector (~)

CSS Combinator Selector

descendant

child

adjacent sibling

general sibling

```
<style>
div p {
  color: red;
}
div > p {
  background-color: yellow;
}
div + p {
  background-color: gray;
}
div ~ p {
  color: blue;
}
</style>
```

CSS Combinator Selector

```
<style>
div p {
  color: red;
}
div > p {
  background-color: yellow;
}
div + p {
  background-color: gray;
}
div ~ p {
  color: blue;
}
</style>
```

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4 in the div.

Paragraph 5. Not in a div.

Paragraph 6. Not in a div.

Paragraph 7. Not in a div.

CSS Pseudo-Class Selector

- Select elements based on state of an element.
  - When a user mouse over it.
  - When a link is visited or unvisited
  - When an element gets focus

- There are many pseudo-classes such as
  - :hover
  - :focus
  - :visited
  - :first-child
  - :disable

CSS Pseudo-Class Selector

- Link

```
<style>
/* unvisited link */
a:link {
  color: red;
}

/* visited link */
a:visited {
  color: green;
}

/* mouse over link */
a:hover {
  color: hotpink;
}

/* selected link */
a:active {
  color: blue;
}
</style>
```

This is a link

This is a link

This is a link

This is a link

CSS Pseudo-Element Selector

- It is used to style part of an element.

- There are five pseudo-elements
  - ::after
  - ::before
  - ::first-letter
  - ::first-line
  - ::selection

CSS Pseudo-Element Selector

```
<style>
p::after {
  content: " - Remember this";
  background-color: yellow;
  color: red;
  font-weight: bold;
}
p::before {
  content: "Read this -";
  color: red;
  font-weight: bold;
}
div::first-letter {
  font-size: 200%;
  color: #8A2BE2;
}
div::first-line {
  background-color: lightgray;
}
::selection {
  color: blue;
  background: hotpink;
}
</style>
```

```
<p>My name is Donald</p>
<div>I live in Ducksburg</div>
```

Read this -My name is Donald **- Remember this**

I live in Ducksburg

CSS Attribute Selector

- Select elements based on attribute.
- There seven attribute selectors
  - [attribute]
  - [attribute=value]
  - [attribute~=value]
  - [attribute|=value]
  - [attribute^=value]
  - [attribute$=value]
  - [attribute*=value]

CSS Attribute Selector

```css
<style>
a[target] {
  background-color: yellow;
}
a[target="_blank"] {
  background-color: lightgray;
}
[title~="work"] {
  color: hotpink;
}
[class^="top"] {
  background: yellow;
}
[class$="test"] {
  background: blue;
}
[class*="se"] {
  background: orange;
}
[class|=top] {
  background: green;
}
</style>
```

```html
<a href="https://www.w3schools.com">w3schools.com</a>
<a href="http://www.disney.com" target="_blank">disney.com</a>
<a href="http://www.wikipedia.org" target="_top">wikipedia.org</a>

<p title="work message">Hello World Message</p>
<p class="top-header">This is paragraph message</p>
<p class="topcontent">Are you learning CSS?</p>
<p class="first_test">The first test element.</p>
<p class="second">The second test element.</p>
```

w3schools.com disney.com wikipedia.org

Hello World Message

This is paragraph message

Are you learning CSS?

The first test element.

The second test element.

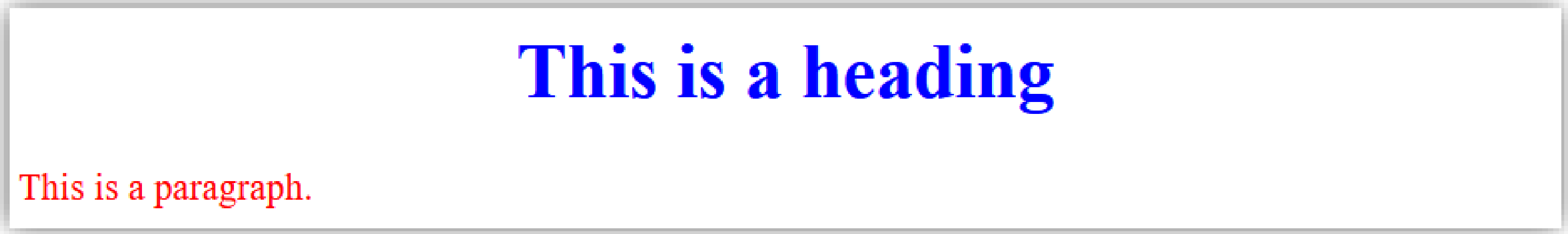
CSS How To

- Inline CSS
- Internal CSS
- External CSS

Inline CSS

- Inline style is used to apply a unique style for a single element by adding style attribute to the element.

```html
<h1 style="color:blue;text-align:center;">This is a heading</h1>
<p style="color:red;">This is a paragraph.</p>
```



Internal CSS

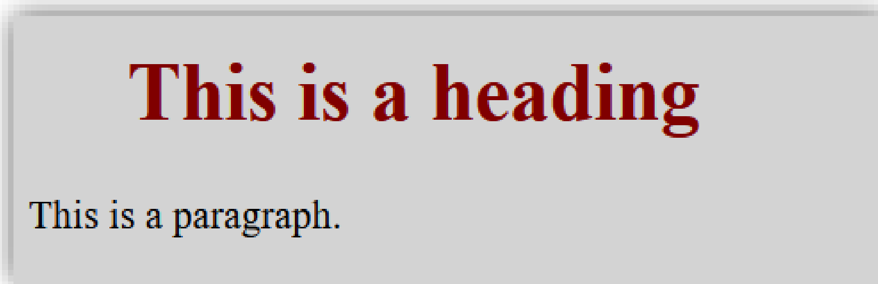
- The internal style is defined inside the <style> element inside the head section.

```html
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-color: lightgray;
}

h1 {
  color: maroon;
  margin-left: 40px;
}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```



External CSS

- The external style sheet is created as a separated file (.css file).
- Each HTML page must include a reference to the style sheet file inside the <link> element within the head section

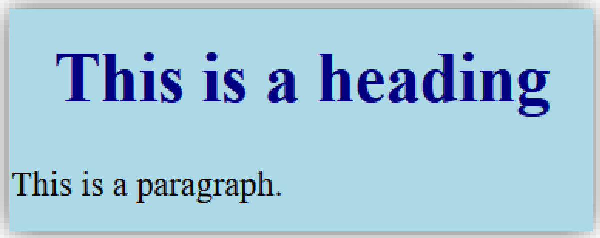
```html
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="mystyle.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

```css
body {
  background-color: lightblue;
}

h1 {
  color: navy;
  margin-left: 20px;
}
```



Exercise 8.1

- Try to style a given webpage which results to similar style as shown in the next slide with three methods of CSS (i.e., inline CSS, internal CSS and external CSS). Submit each method to the LEB2 assignment by means of zip file.
- Hints: using only these properties should be enough:
  - background-color
  - color
  - text-align

CSS Exercise

File | D:/GoogleDrive/SIT/BIS605/Mockupweb/exercise8_solution.html

School of Information Technology

Doctoral Programs

The Doctor of Philosophy programs aim to produce highly qualified researchers and scholars to international standards. It also aims to create researchers of integrity as clearly stated in the SIT "Five P's" philosophy: Prompt actions, Punctuality, Professionalism, Perseverance, and Principles and can respond to computer science industry requirements. Moreover, the program aims for the discovery of new insights and academic excellence for national development by emphasizing in-depth research and the use of resources at maximum potential.

SIT provides two doctoral programs as follows:

- Information Techonology
- Computer Science

Information Technology

Philosophy
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Program objectives:
1. To produce productive researchers and scholars in information technology to international standards and promote significant human resources for the betterment of society
2. To conduct original research, develop new technology in information technology, and collaborate academically with leading foreign universities
3. To achieve for an academic research excellence, not only within Thailand but also abroad through continuous research development

Computer Science

Philosophy
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Program objectives:
1. To produce productive researchers and scholars in computer science to international standards and promote significant human resources for the betterment of society
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CSS PROPERTIES

CSS Background

- Some well-known background properties
  - background-color
  - background-image
  - background-attachment
  - background-repeat
  - background-position
  - background

CSS Background

- background-color – set background color of an element
- opacity – specifues the opacity or transparency of an element with the value range of 0.0-1.0

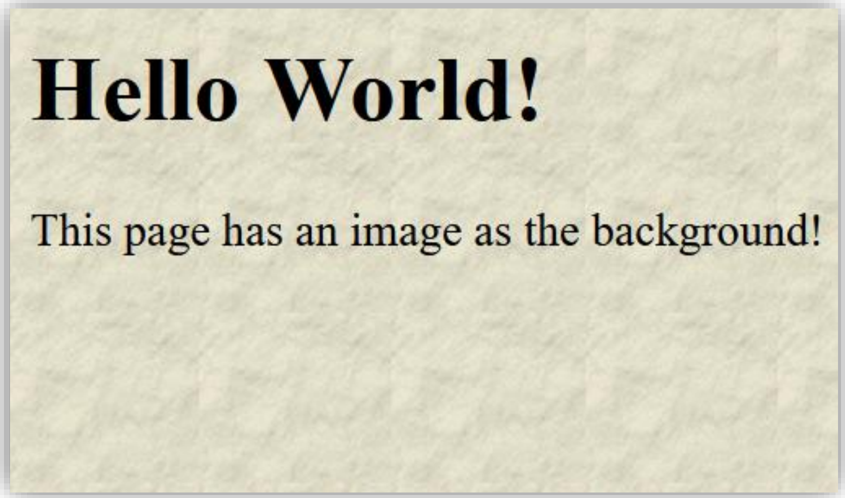
```
<head>
<style>
p {
  background-color: pink;
  opacity: 0.5
}
</style>
</head>
<body>
<p>The international relations (IR) team
at SIT is here to help both SIT students
interested in studying abroad and exchange
students from our exchange partners. We
are the first point of contact for these
students and will guide them through the
application process, in partnership with
the university @ KMUTT International
Affairs.
</p>
</body>
```

The international relations (IR) team at SIT is here to help both SIT students interested in studying abroad and exchange students from our exchange partners. We are the first point of contact for these students and will guide them through the application process, in partnership with the university @ KMUTT International Affairs.

CSS Background

- background-image – sets background image of an element. The image will be repeat by default.

```
<head>
<style>
body {
  background-image: url("paper.gif");
}
</style>
</head>
<body>
<h1>Hello World!</h1>
<p>This page has an image as the background!</p>
</body>
```

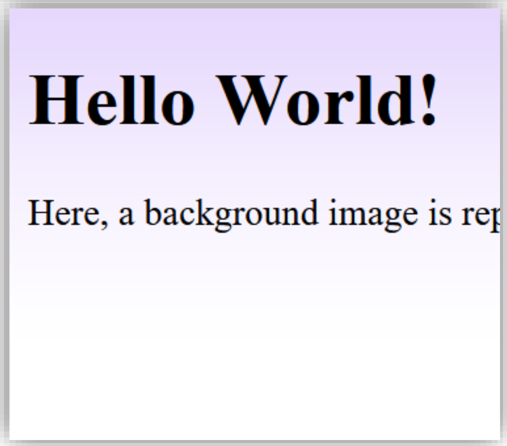


CSS Background

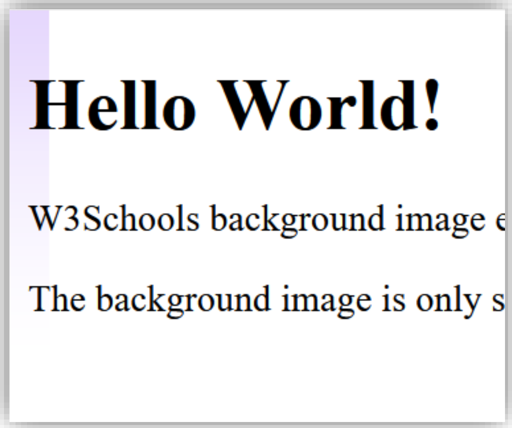
- background-repeat – sets the property of background if it should repeat both horizontally and vertically.
- The value can be: no-repeat, repeat-x (repeat horizontally), and repeat-y (repeat vertically)

```css
body {
  background-image: url("gradient_bg.png");
  background-repeat: repeat-y;
}
```


```css
body {
  background-image: url("gradient_bg.png");
  background-repeat: repeat-x;
}
```



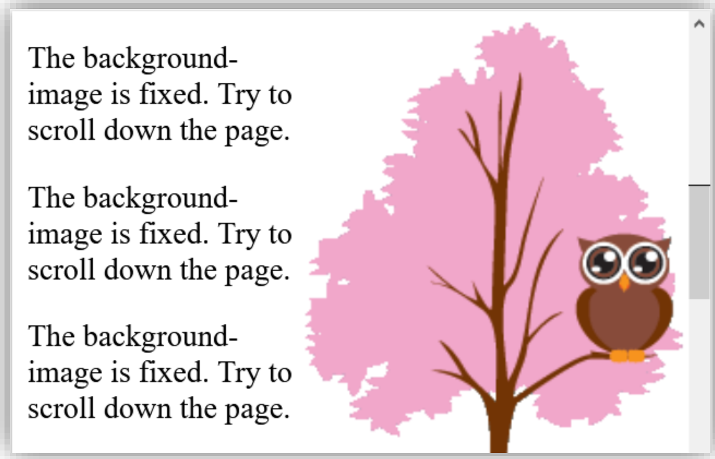

```css
body {
  background-image: url("gradient_bg.png");
  background-repeat: no-repeat;
}
```



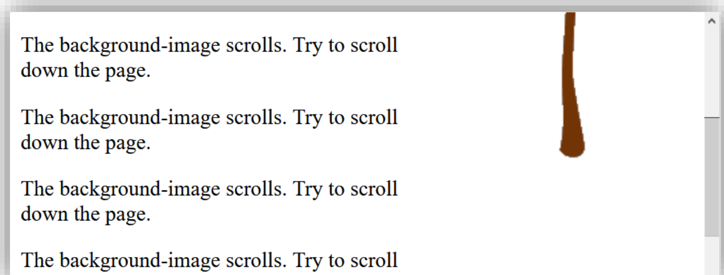
CSS Background

- background-attachment sets the property of background if it should scroll or be fixed.

```
body {
  background-image: url("img_tree.png");
  background-repeat: no-repeat;
  background-position: right top;
  margin-right: 200px;
  background-attachment: fixed;
}
```



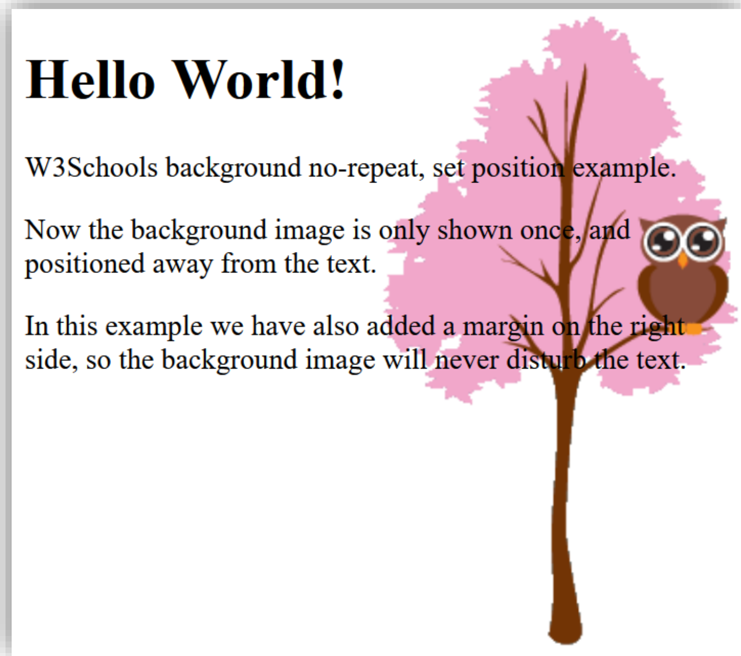

```
body {
  background-image: url("img_tree.png");
  background-repeat: no-repeat;
  background-position: right top;
  margin-right: 200px;
  background-attachment: scroll;
}
```



CSS Background

- background-position sets the position of background.
  - top
  - bottom
  - left
  - right
  - center
  - top left, top right
  - bottom left, bottom right
  - position of x and y (percent or length)

```
body {
  background-image: url("img_tree.png");
  background-repeat: no-repeat;
  background-position: right top;
}
```



CSS Background

- background – a shorthand property for setting background properties at once. The order of values is:
  - background-color
  - background-image
  - background-repeat
  - background-attachment
  - background-position

```css
body {
  background-color: #ffffff;
  background-image: url("img_tree.png");
  background-repeat: no-repeat;
  background-position: right top;
}
```

```css
body {
  background: #ffffff url("img_tree.png") no-repeat right top;
}
```

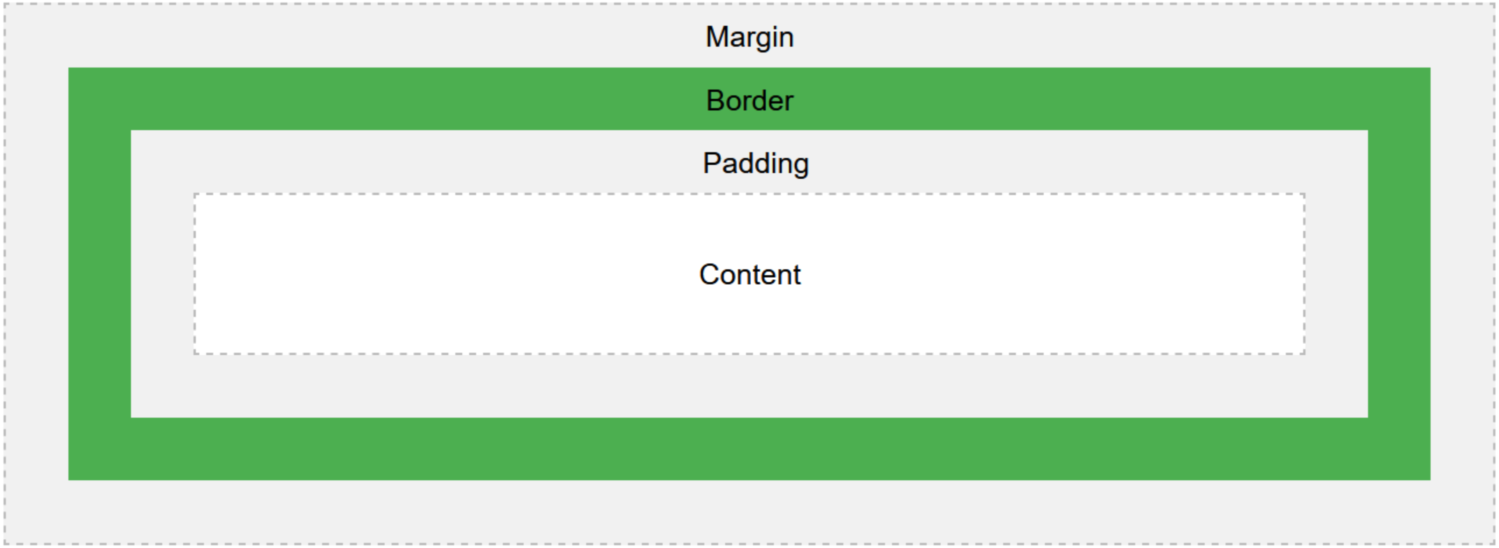
CSS Height and Width

- Each element has a width of 100%.
- The property of height and width are used to specify height and width of an element which can have below values:
  - auto – default (browser will calculate itself)
  - length – height/width in specific units such as px (relative to device), in, cm, mm etc.
  - %
  - initial – default value
  - inherit – same with parent

```
div {
  height: 200px;
  width: 50%;
  background-color: powderblue;
}
```

CSS Box Model

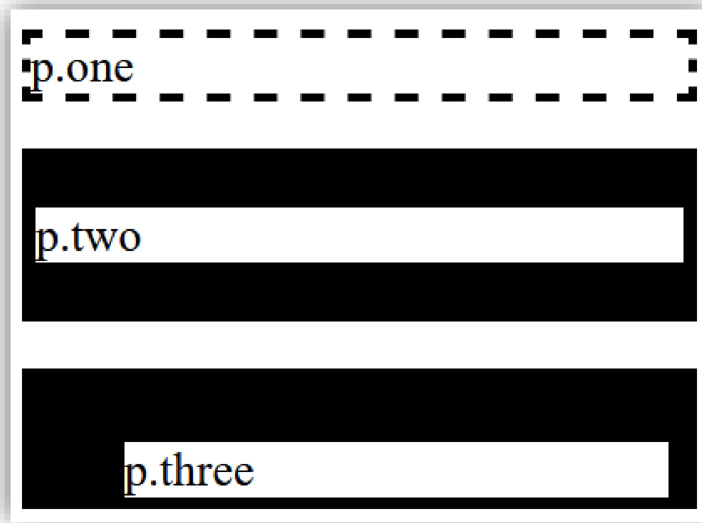
- HTML elements are designed as a box. In CSS, we use box model to consider when design an element.



CSS Border

- border-style – specifies type of border which can be dotted, dashed, solid, double, etc.
- border-width – specifies thickness of borders. It can be defined using specific size (px, cm, in etc.) or pre-defined values: thin, medium, or thick.

```css
p.one {
  border-style: dashed;
  border-width: medium;
}
p.two {
  border-style: solid;
  border-width: 20px 5px;
  /* 20px top and bottom, 5px on the sides */
}
p.three {
  border-style: solid;
  border-width: 25px 10px 4px 35px;
  /* 25px top, 10px right, 4px bottom and 35px left */
}
```



CSS Border

- To specify style of each border side, CSS provide specific properties:
  - border-top-style
  - border-right-style
  - border-bottom-style
  - border-left-style

```
p {
  border-top-style: dotted;
  border-right-style: solid;
  border-bottom-style: dotted;
  border-left-style: solid;
}
```

Different Border Styles

CSS Border

- border-color – specifies color of borders.
- border-radius – specifies rounded borders

```
p.one {
  border-style: dashed;
  border-width: medium;
  border-color: red;
}
p.two {
  border-style: solid;
  border-width: 20px 5px;
  border-color: green;
  border-radius: 5px;
}
p.three {
  border-style: solid;
  border-width: 25px 10px 4px 35px;
  border-color: blue;
  border-radius: 50px;
}
```

CSS Border

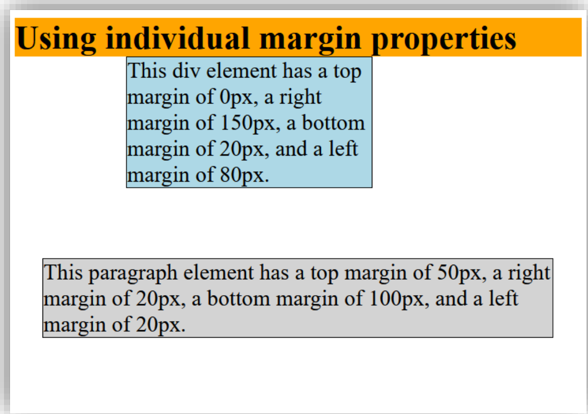
- border shorthand property can help to specify three properties at once:
  - border-width
  - border-style (required)
  - border-color

```css
p {
  border: 5px solid red;
}
```

CSS Margins

- We can control space around element (outside border) by using margin property. There are four specific margins:
  - margin-top
  - margin-right
  - margin-bottom
  - margin-left
  - margin (top, right, bottom, left)

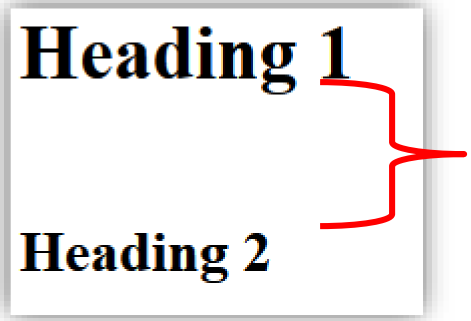
```css
h2 {
    background-color: orange;
    margin-bottom: 0px;
}
div {
  border: 1px solid black;
  margin-top: 0px;
  margin-bottom: 20px;
  margin-right: 150px;
  margin-left: 80px;
  background-color: lightblue;
}
p {
  border: 1px solid black;
  margin: 50px 20px 100px 20px;
  background-color: lightgray;
}
```



CSS Margins

- Margins on top and bottom are sometimes collapsed into a single margin. The collapsed margin will equal to the largest margin of the two margins.
- This does not happen on left and right margins!

```
h1 {
  margin: 0 0 50px 0;
}

h2 {
  margin: 20px 0 0 0;
}
```



This margins should be 70px, but the actual margin is collapsed to 50px.

CSS Padding

- We can control space around element’s content (inside border) by using margin property. There are four specific margins:
  - padding-top
  - padding-right
  - padding-bottom
  - padding-left
  - padding (top, right, bottom, left / top, right, bottom / top&bottom, right&left / all)

```css
div {
  border: 1px solid black;
  padding: 25px 50px 75px 100px;
  background-color: lightblue;
}
```

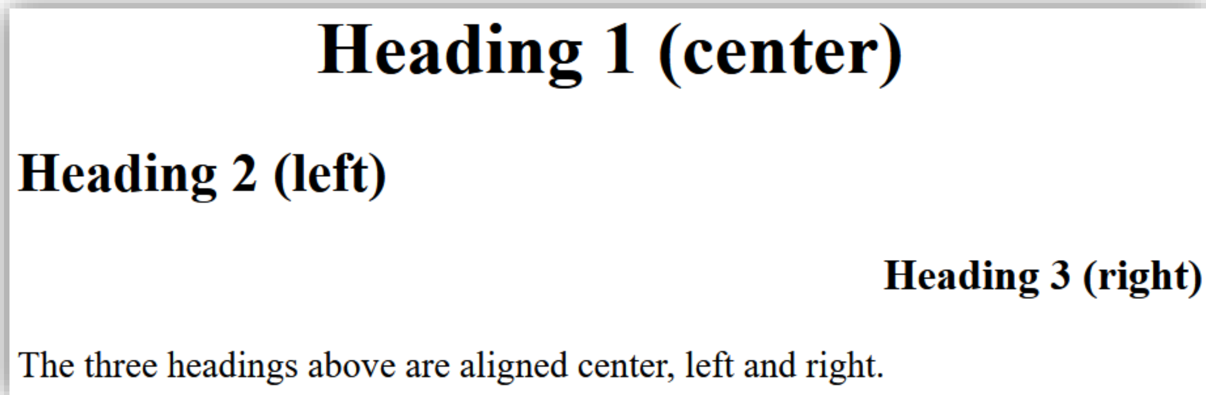
> This div element has a top padding of 25px, a right padding of 50px, a bottom padding of 75px, and a left padding of 100px.

CSS Text & Font

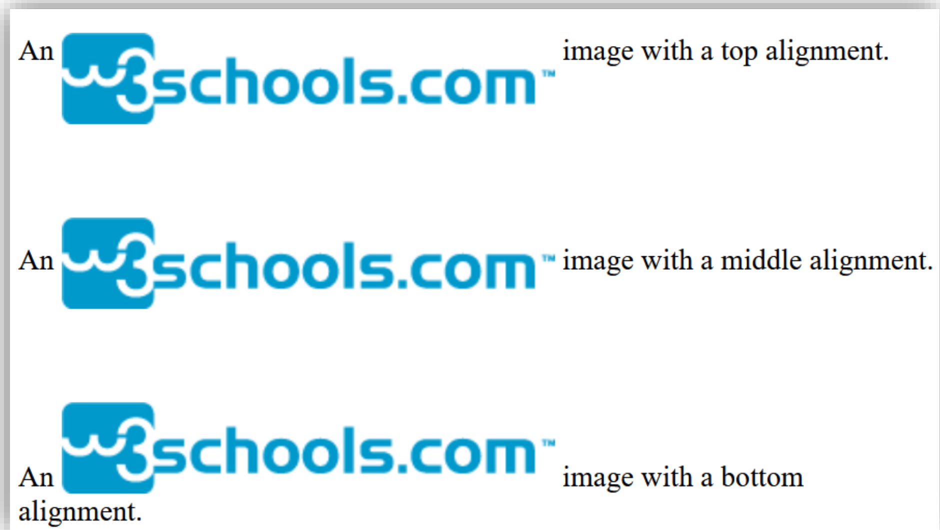
- Text can be formatted using many CSS properties:
  - color – sets text color
  - background-color – sets background color of text
  - text-align – sets horizontal alignment of a text (and other elements)
    - center
    - left
    - right
    - justify
  - vertical-align – set vertical alignment of a text (and other elements)
    - top
    - middle
    - bottom

CSS Text & Font

```css
h1 {
  text-align: center;
}

h2 {
  text-align: left;
}

h3 {
  text-align: right;
}
```




```css
img.top {
  vertical-align: top;
}

img.middle {
  vertical-align: middle;
}

img.bottom {
  vertical-align: bottom;
}
```



CSS Text & Font

- font-family – sets font family of text. If font name contain more than one word, it should be used within quotation mark.

```css
.serif {
  font-family: "Times New Roman", Times, serif;
}

.sansserif {
  font-family: Arial, Helvetica, sans-serif;
}

.monospace {
  font-family: "Lucida Console", Courier, monospace;
}
```

This is a paragraph, shown in the Times New Roman font.

This is a paragraph, shown in the Arial font.

This is a paragraph, shown in the Lucida Console font.

CSS Text & Font

- text-indent – specifies indentation of the first line of a text.
- letter-spacing – specifies the space between the characters in a text.
- line-height – is used to define space between lines.
- word-spacing – is used to define space between words in a text.
- white-space – specifies how white space inside an element is handled.

```
p {
  text-indent: 50px;
  letter-spacing: 3px;
  line-height: 1.8;
  word-spacing: 10px;
  /*white-space: nowrap;*/
}
```

In my younger and more vulnerable years my father gave me some advice that I've been turning over in my mind ever since. 'Whenever you feel like criticizing anyone,' he told me, 'just remember that all the people in this world haven't had the advantages that you've had.'

```
p {
  text-indent: 50px;
  letter-spacing: 3px;
  line-height: 1.8;
  word-spacing: 10px;
  white-space: nowrap;
}
```

In my younger and

CSS Float

- float property specifies how an element should float.
  - left – floats to the left of its container
  - right – floats to the right of its container
  - none – not float (default)
  - inherit – inherit float property of its parent
- clear property specifies what elements can float beside the cleared element and on which side.
  - none – allows floating elements on both side (default)
  - left – no floating elements on left side
  - right – no floating elements on right side
  - both – no floating elements on either left or right
  - inherit – inherits clear property from its parent

CSS Float

```css
img {
  float: right;
}
```

```css
img {
  float: none;
}
```

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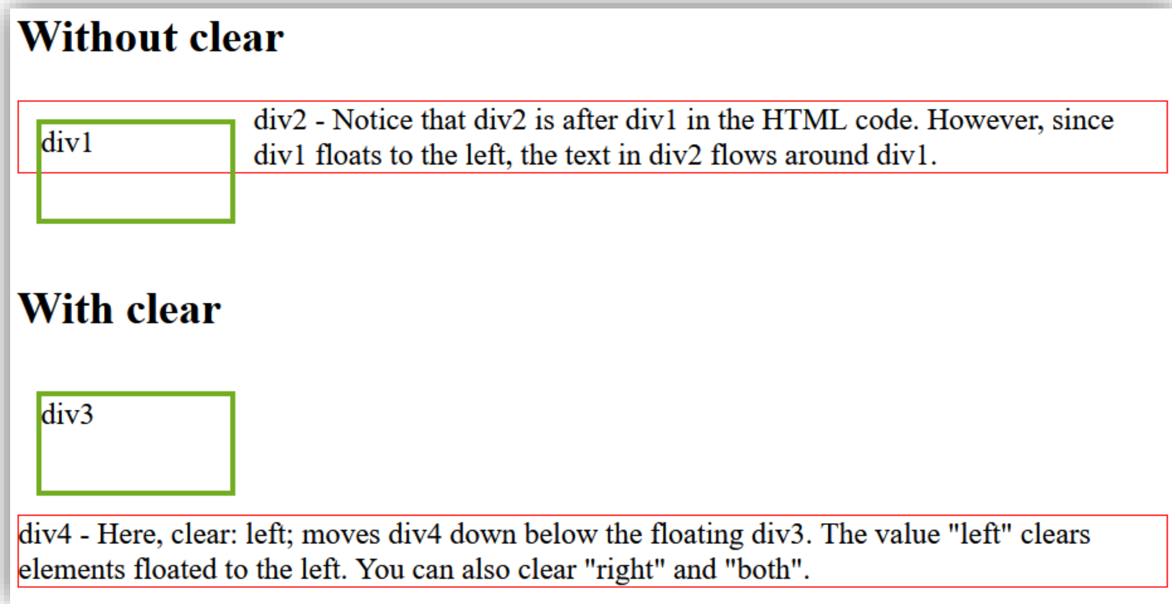
CSS Float

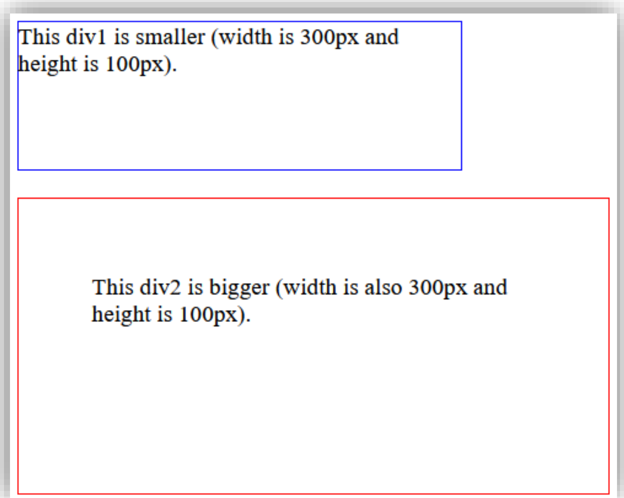
```
.div1 {
  float: left;
  width: 100px;
  height: 50px;
  margin: 10px;
  border: 3px solid #73AD21;
}

.div2 {
  border: 1px solid red;
}

.div3 {
  float: left;
  width: 100px;
  height: 50px;
  margin: 10px;
  border: 3px solid #73AD21;
}

.div4 {
  border: 1px solid red;
  clear: left;
}
```



CSS Box Sizing

- box-sizing – allows us to include the padding and border in an element's total width and height.
- By default, the calculation of actual size of each element will follow these:
  - width + padding + border = actual width
  - height + padding + border = actual height

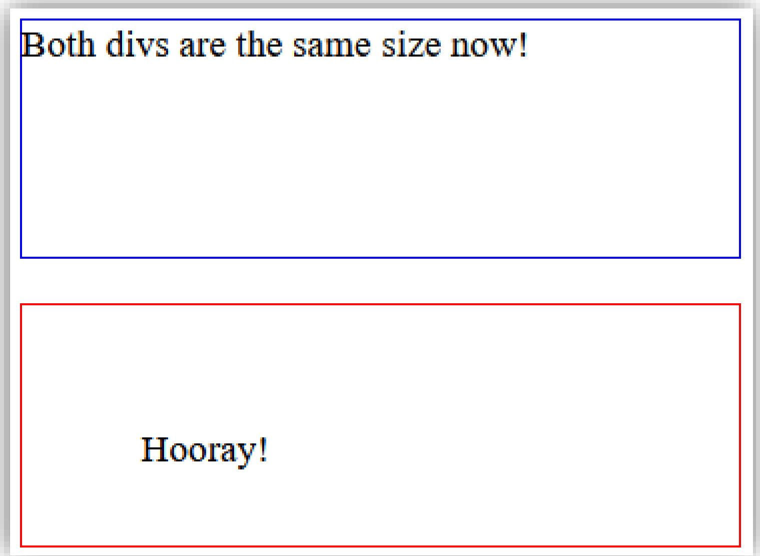
```css
.div1 {
  width: 300px;
  height: 100px;
  border: 1px solid blue;
}

.div2 {
  width: 300px;
  height: 100px;
  padding: 50px;
  border: 1px solid red;
}
```



CSS Box Sizing

- With the box-sizing property will help us include all padding and border in an element’s total width and height.

```
.div1 {
  width: 300px;
  height: 100px;
  border: 1px solid blue;
  box-sizing: border-box;
}

.div2 {
  width: 300px;
  height: 100px;
  padding: 50px;
  border: 1px solid red;
  box-sizing: border-box;
}
```



CSS RESPONSIVE

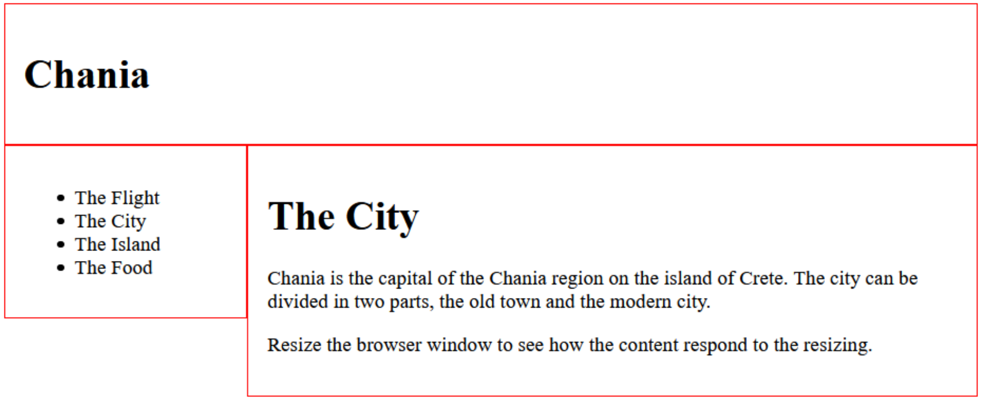
Viewport

- The viewport is the user's visible area of a web page.
- The viewport varies with the device.
- To set viewport, we use <meta> tag

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

CSS Grid View

```html
<div class="header">
  <h1>Chania</h1>
</div>

<div class="row">

<div class="col-3">
  <ul>
    <li>The Flight</li>
    <li>The City</li>
    <li>The Island</li>
    <li>The Food</li>
  </ul>
</div>

<div class="col-9">
  <h1>The City</h1>
  <p>Chania is the capital of the Chania region on the island of Crete. The city can be
divided in two parts, the old town and the modern city.</p>
  <p>Resize the browser window to see how the content respond to the resizing.</p>
</div>

</div>
```

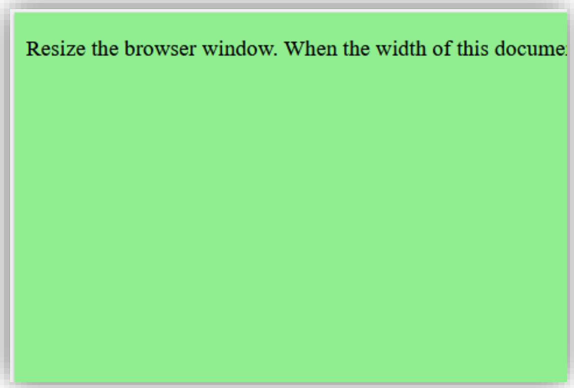


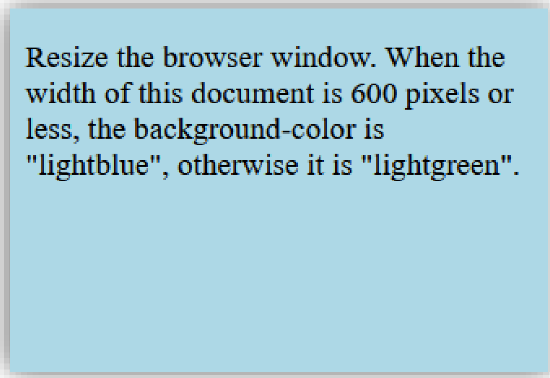

```css
* {
  box-sizing: border-box;
}

.header {
  border: 1px solid red;
  padding: 15px;
}

.row::after {
  content: "";
  clear: both;
  display: table;
}

[class*="col-"] {
  float: left;
  padding: 15px;
  border: 1px solid red;
}

.col-1 {width: 8.33%;}
.col-2 {width: 16.66%;}
.col-3 {width: 25%;}
.col-4 {width: 33.33%;}
.col-5 {width: 41.66%;}
.col-6 {width: 50%;}
.col-7 {width: 58.33%;}
.col-8 {width: 66.66%;}
.col-9 {width: 75%;}
.col-10 {width: 83.33%;}
.col-11 {width: 91.66%;}
.col-12 {width: 100%;}
```

Media Query

- We use @media rule to include a block of CSS properties only if a certain condition is true.

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width,
initial-scale=1.0">
<style>
body {
  background-color: lightgreen;
}

@media only screen and (max-width: 600px) {
  body {
    background-color: lightblue;
  }
}
</style>
</head>
<body>

<p>Resize the browser window. When the width of this
document is 600 pixels or less, the background-color
is "lightblue", otherwise it is "lightgreen".</p>

</body>
</html>
```





Media Query

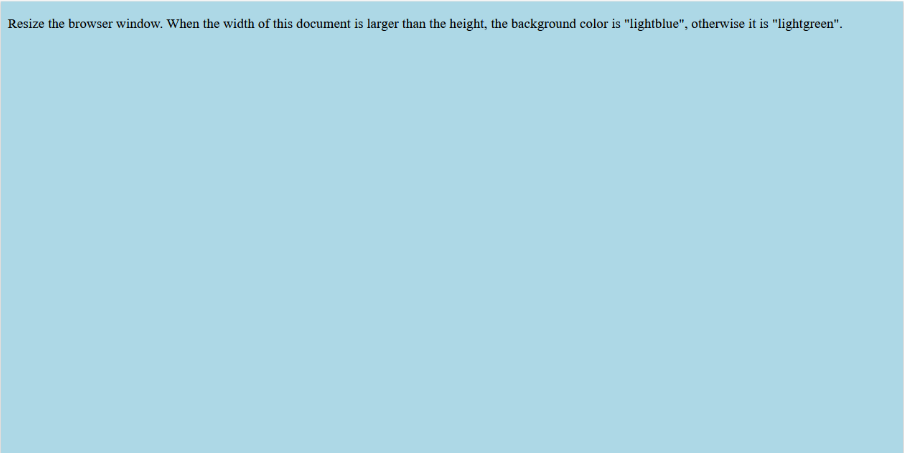
- Typical breakpoints

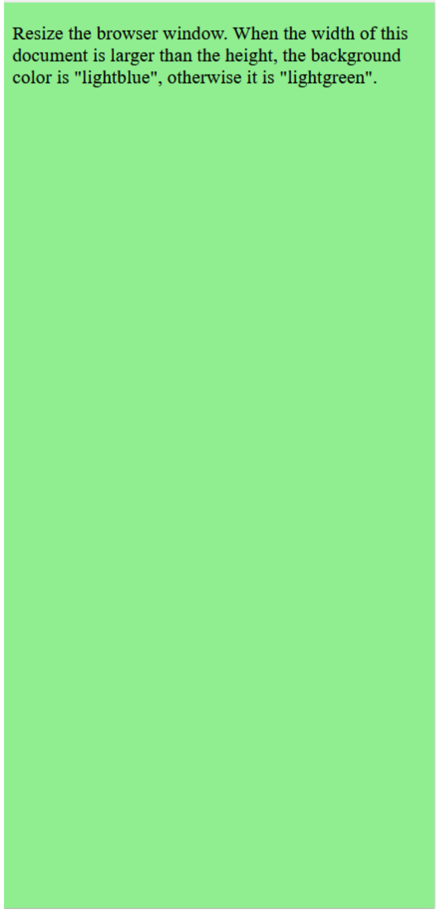
```css
/* Extra small devices (phones, 600px and down) */
@media only screen and (max-width: 600px) {...}

/* Small devices (portrait tablets and large phones, 600px and up) */
@media only screen and (min-width: 600px) {...}

/* Medium devices (landscape tablets, 768px and up) */
@media only screen and (min-width: 768px) {...}

/* Large devices (laptops/desktops, 992px and up) */
@media only screen and (min-width: 992px) {...}

/* Extra large devices (large laptops and desktops, 1200px and up) */
@media only screen and (min-width: 1200px) {...}
```

Media Query

- Media query can be used to change layout as well.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<style>
body {
  background-color: lightgreen;
}

@media only screen and (orientation: landscape) {
  body {
    background-color: lightblue;
  }
}
</style>
```



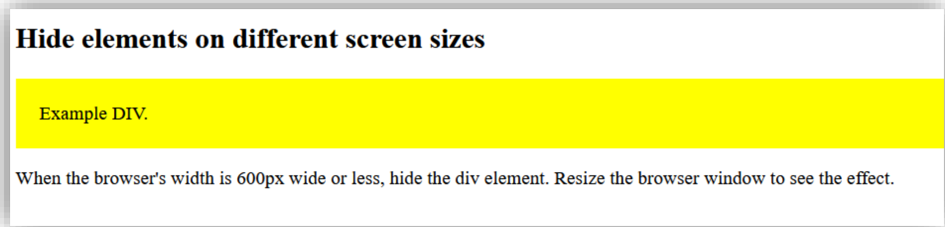


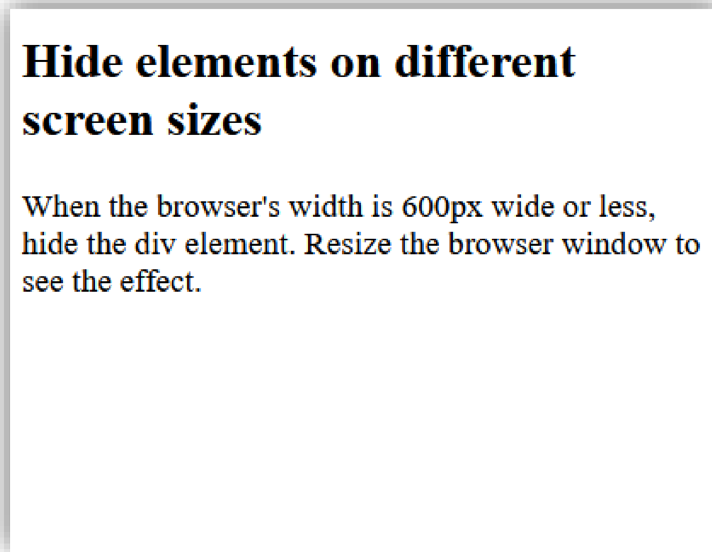
Media Query

- Media query can be used to hide some elements when screen size is different.

```css
div.example {
  background-color: yellow;
  padding: 20px;
}

@media screen and (max-width: 600px) {
  div.example {
    display: none;
  }
}
```

```html
<h2>Hide elements on different screen sizes</h2>

<div class="example">Example DIV.</div>

<p>When the browser's width is 600px wide or less,
hide the div element. Resize the browser window to
see the effect.</p>
```





CSS Max-width

- Using the following CSS of width and height can make images and video responsive.

```css
img {
  width: 100%;
  height: auto;
}
```

- If we want to control the maximum size of the image/video, we can use max-width property.

```css
img {
  max-width: 100%;
  height: auto;
}
```

Exercise 8.2

- From the previous exercise (8.1), try to:
  - Modify the layout so that the <article> of Information Technology and Computer Science located side-by-side
  - Use media query if the screen is portrait (orientation: portrait) i.e., the height is larger than the width, make the <article> disappear.

CSS Exercise

File | D:/GoogleDrive/SIT/BIS605/Mockupweb/exercise8_solution.html

School of Information Technology

Doctoral Programs

The Doctor of Philosophy programs aim to produce highly qualified researchers and scholars to international standards. It also aims to create researchers of integrity as clearly stated in the SIT "Five P's" philosophy: Prompt actions, Punctuality, Professionalism, Perseverance, and Principles and can respond to computer science industry requirements. Moreover, the program aims for the discovery of new insights and academic excellence for national development by emphasizing in-depth research and the use of resources at maximum potential.

SIT provides two doctoral programs as follows:

- Information Techonology
- Computer Science

Information Technology

Philosophy
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Program objectives:
1. To produce productive researchers and scholars in information technology to international standards and promote significant human resources for the betterment of society
2. To conduct original research, develop new technology in information technology, and collaborate academically with leading foreign universities
3. To achieve for an academic research excellence, not only within Thailand but also abroad through continuous research development

Computer Science

Philosophy
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CSS Exercise

File | D:/GoogleDrive/SIT/...

School of Information Technology

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